Guidelines for Mod position scoring

```
<SpectrumIdentificationProtocol analysisSoftware_ref="ID_software" id="SearchProtocol_1">
  <SearchType>
    <cvParam accession="MS:1001083" cvRef="PSI-MS" name="ms-ms search"/>
  </SearchType>
  <AdditionalSearchParams>
    <cvParam accession="MS:1001211" cvRef="PSI-MS" name="parent mass type mono"/>
    <cvParam accession="MS:1001256" cvRef="PSI-MS" name="fragment mass type mono"/>
    <cvParam accession="MS:1002491" cvRef="PSI-MS" name="Modification localization scoring"/>
  </AdditionalSearchParams>
  ....
  <Threshold>
    <cvParam accession="MS:1002354" cvRef="PSI-MS" name="PSM-level q-value" value="0.01"/>
    <cvParam accession="MS:1002380" cvRef="PSI-MS" name="false localization rate" value="0.05"/>
  </Threshold>
```

A

B

```
<Peptide id="KYYGNVVYYIGER_p@2|3">
  <PeptideSequence>KYYGNVVYYIGER</PeptideSequence>
  <Modification monoisotopicMassDelta="79.966331" location="2" residues="Y">
    <cvParam accession="UNIMOD:21" cvRef="UNIMOD" name="Phospho"/>
    <cvParam accession="MS:1002504" cvRef="PSI-MS" name="modification index" value="1"/>   </Modification>
</Peptide>
<Peptide id="KYYGNVVYYIGER_p@2|3_p@8|9">
  <PeptideSequence>KYYGNVVYYIGER</PeptideSequence>
  <Modification monoisotopicMassDelta="79.966331" location="2" residues="Y">
    <cvParam accession="UNIMOD:21" cvRef="UNIMOD" name="Phospho"/>
    <cvParam accession="MS:1002504" cvRef="PSI-MS" name="modification index" value="1"/>
  </Modification>
  <Modification monoisotopicMassDelta="79.966331" location="8" residues="Y">
    <cvParam accession="UNIMOD:21" cvRef="UNIMOD" name="Phospho"/>
    <cvParam accession="MS:1002504" cvRef="PSI-MS" name="modification index" value="2"/>
  </Modification>
</Peptide>
```

C

D

```
<SpectrumIdentificationResult spectraData_ref="qExactive01819.mgf" spectrumID="index=2727" id="SIR_4207">
  <SpectrumIdentificationItem passThreshold="true" rank="1"  peptide_ref="DNSTMGYMMAK_15.99491461956_15.99491461956"
    calculatedMassToCharge="640.751423992447"  experimentalMassToCharge="640.751992494115"
    chargeState="2" id="SII_4207_1">
   <PeptideEvidenceRef peptideEvidence_ref="PepEv_9145"/>
   <cvParam cvRef="PSI-MS" accession="MS:1001969" name="phosphoRS score" value="1:66.66666666:5:false"/>
   <cvParam cvRef="PSI-MS" accession="MS:1001969" name="phosphoRS score" value="1:66.66666666:8:false"/>
   <cvParam cvRef="PSI-MS" accession="MS:1001969" name="phosphoRS score" value="1:66.66666666:9:false"/>

   <cvParam cvRef="PSI-MS" accession="MS:1002539" name="D-score" value="1:62.3758727298785 8:5:false"/>
   <cvParam cvRef="PSI-MS" accession="MS:1002539" name="D-score" value="1:14.141586139372523:8:false"/>
   <cvParam cvRef="PSI-MS" accession="MS:1002539" name="D-score" value="1:36.626793239964236:9:false"/>
   <!-- Other PSM-level scores not shown... -->
  </SpectrumIdentificationItem>
  <!-- Other ranked identifications not shown... -->
</SpectrumIdentificationResult>
```

E

F

Feature	Explanation
A	If modification rescoring has been performed, this cvParam MUST be present.
B	A Threshold for modification localizations MAY be inserted into the <SpectrumIdentificationProtocol>
C	The ambiguity with respect to modification location is present at the level of <SpectrumIdentificationItem> but rescored software SHOULD use the residues and location attribute to insert the most likely location for the modification
D	If Feature A is present, every <Modification> element MUST have the cvParam used as a unique identifier to be referenced by Feature F.
E	If Feature A is present, every <SpectrumIdentificationItem> referencing a peptide with a variable modification MUST have a cvParam for the location score. The value slot takes the following format MOD_INDEX:SCORE:POSITION:PASS_THRESHOLD **MOD_INDEX** = <Modification> index attribute in the referenced <Peptide> object **SCORE** = Score or statistical measure associated with the modification position **POSITION** = Position of the modification on the peptide (N-terminus = 0, C-terminus = peptide length + 1). If the score pertains to grouped positions, different positions MUST be separated by "|" **PASS_THRESHOLD** = true | false with regards to the threshold specified in Feature A. If no Threshold has been specified, this MUST always be true.
F	The modification position rescoring software SHOULD NOT include additional equal or lower ranked <SpectrumIdentificationItem> elements referencing a different <Peptide> element with the same peptide sequence and the same set of modifications (but with different modification positions) i.e. the only expected mechanism for specifying modification position is the cvParam specified in Feature D.